INVARIANT RANDOM C^*-SUBALGEBRAS OF PRODUCTS OF HYPERBOLIC GROUPS

Abstract. We present a proof of boomerang rigidity for finite products of torsion-free nonamenable hyperbolic groups, extending the single-factor theorem of Fima–Oppelmayer–Tao. Every boomerang unital subalgebra of the reduced group C^*-algebra is a subgroup algebra. Consequently, the subgroup-algebra correspondence induces an affine bijection between invariant random subgroups and invariant random C^*-subalgebras. The proof combines the selective averaging of Amrutam–Jiang with recurrence in the space of subalgebras. Coordinate expectations remove terms with a trivial active coordinate; a common return sequence then isolates each nonzero Fourier coefficient. The argument also applies to countable restricted direct sums.

1. Introduction

Let Γ be a countable discrete group. Conjugation by the left regular representation defines an action on the space $\mathrm{Sub}(C_r^*(\Gamma))$ of unital C^*-subalgebras containing the ambient unit. We equip this space with the Wijsman topology. An invariant random C^*-subalgebra is an invariant Borel probability measure on this space. Every invariant random subgroup gives such a measure through

$$\iota_r : \mathrm{Sub}(\Gamma) \longrightarrow \mathrm{Sub}(C_r^*(\Gamma)), \qquad H \longmapsto C_r^*(H).$$

Following Fima–Oppelmayer–Tao, a random subalgebra obtained in this way is called *geometric*.

Fima–Oppelmayer–Tao prove that every invariant random subalgebra is geometric when Γ is torsion-free, nonamenable, and hyperbolic [FOT26, Corollary 6.9]. Their stronger statement classifies boomerang subalgebras, whose conjugates return to the original algebra along a sequence of positive powers of each group element [FOT26, Theorem 6.8]. In the deterministic setting, Amrutam–Jiang prove that every invariant unital subalgebra of $C_r^*(\Gamma_1 \times \cdots \times \Gamma_n)$ is a normal subgroup algebra when the factors have these hypotheses [AJ26, Theorem 5.4].

This paper extends the boomerang and random conclusions to finite products of the same class of groups. The main statement is as follows.

Theorem A. *Let $\Gamma = \Gamma_1 \times \cdots \times \Gamma_n$, where $n \geq 1$ and each Γ_i is a torsion-free nonamenable hyperbolic group. If $B \in \mathrm{Sub}(C_r^*(\Gamma))$ is boomerang, then*

$$B = C_r^*(H_B), \qquad H_B = \{g \in \Gamma : \lambda_g \in B\}.$$

The probabilistic consequence includes a precise inverse to the subgroup-algebra correspondence.

Corollary B. *For Γ as in Theorem A, every invariant random C^*-subalgebra is geometric. The map*

$$\mathrm{IRS}(\Gamma) \longrightarrow \mathrm{IRA}(C_r^*(\Gamma)), \qquad \eta \longmapsto (\iota_r)_*\eta,$$

is an affine bijection, with inverse $\nu \mapsto (B \mapsto H_B)_*\nu$. It preserves and reflects ergodicity.

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In particular, the conclusion applies to $\mathbb{F}_2 \times \mathbb{F}_2$. The subgroup H_B need not split across the coordinates. For example, if $\chi : \mathbb{F}_2 \to \mathbb{Z}$ is onto, then

$$H = \{(g,h) \in \mathbb{F}_2 \times \mathbb{F}_2 : \chi(g) = \chi(h)\}$$

is normal. If $\chi(s) = 1$, then $\lambda_{(s,s)} \in C_r^*(H)$, whereas $\lambda_{(s,e)}, \lambda_{(e,s)} \notin C_r^*(H)$. Thus the proof must recover a tuple unitary without requiring its individual coordinate unitaries to belong to the algebra.

The proof follows the coefficient argument in [FOT26, Lemmas 6.5–6.7 and Theorem 6.8]. The product argument requires two additional steps. First, simultaneous selective averaging [AJ26, Theorem 3.3] shows that coordinate cyclic expectations preserve a boomerang algebra. Composing two such expectations gives the coordinate trace map. Second, these trace maps remove terms that are trivial in any nontrivial coordinate of a selected group element. The reduced-word calculation then isolates the selected coefficient, using one return sequence for a tuple of conjugating elements.

2. Notation and preliminaries

We use the notation of [FOT26] for subalgebras and subgroup expectations. Put $A = C_r^*(\Gamma)$, and let λ_g denote the left regular unitary associated with $g \in \Gamma$. The canonical trace is τ_0, and

$$\widehat{a}(g) = \tau_0(a\lambda_g^*) \qquad (a \in A,\ g \in \Gamma).$$

For $B \in \mathrm{Sub}(A)$, write

$$d(a,B) = \inf_{b \in B} \|a - b\|.$$

The Wijsman topology is the weakest topology making all these distance functions continuous. The space $\mathrm{Sub}(A)$ is Polish when A is separable [FOT26, Theorem 2.3].

For $H \leq \Gamma$, let $E_H : A \to C_r^*(H)$ be the canonical conditional expectation. Thus

$$E_H(\lambda_g) = 1_H(g)\lambda_g.$$

It is unital completely positive and contractive. For subgroups H, K,

$$E_H E_K = E_K E_H = E_{H \cap K}. \tag{2.1}$$

This follows on group unitaries and then by norm density.

Write $\alpha_g = \mathrm{Ad}(\lambda_g)$. Each α_g is a norm isometry and induces a homeomorphism of $\mathrm{Sub}(A)$. A subalgebra B is *boomerang* if, for every $s \in \Gamma$, there are strictly increasing positive integers m_j such that $\alpha_{s^{m_j}}(B) \to B$ in the Wijsman topology. We call (m_j) a return sequence for s at B. For every $a \in B$ and such a sequence,

$$d(\alpha_{s^{-m_j}}(a), B) = d(a, \alpha_{s^{m_j}}(B)) \longrightarrow 0. \tag{2.2}$$

We recall two elementary distance estimates, also underlying [FOT26, Lemma 6.7]. If $T : A \to A$ is a linear contraction with $T(B) \subseteq B$, then

$$d(Ta, B) \leq d(a, B). \tag{2.3}$$

For $u, v \in A$,

$$d(uv, B) \leq \|u\| d(v, B) + \|v\| d(u, B) + d(u, B) d(v, B). \tag{2.4}$$

Indeed, choose $b, c \in B$ close to u, v and use $uv - bc = (u - b)v + b(v - c)$, together with $\|b\| \leq \|u\| + \|u - b\|$.

An element $t \neq e$ is called *primitive* if $t = u^k$ implies $k = \pm 1$. In a torsion-free nonamenable hyperbolic group, every nonidentity element has a primitive root, and its centralizer is infinite cyclic with a primitive generator; see [AJ26, Remark 4.2] and the proof of [FOT26, Lemma 6.5].

Every primitive element has a free partner: there is h for which $\langle t, h\rangle$ is free on t, h [AD19, Theorem 0.2].

For products, all tensor products are minimal. Let τ_0^i be the canonical trace on $C_r^*(\Gamma_i)$. Following [AJ26, Section 5.2], denote the coordinate trace expectation by $\mathbb{E}_{\tau_0^i}$. On elementary tensors,

$$\mathbb{E}_{\tau_0^i}(a_1 \otimes \cdots \otimes a_n) = a_1 \otimes \cdots \otimes (\tau_0^i(a_i)1) \otimes \cdots \otimes a_n.$$

For a primitive $t \in \Gamma_i$, we also use the coordinate notation

$$E_{\langle t\rangle}^{(i)} = \mathrm{id} \otimes \cdots \otimes E_{\langle t\rangle} \otimes \cdots \otimes \mathrm{id}\,.$$

The latter notation denotes the lift of the subgroup expectation in factor i.

3. Coordinate expectations

We use the following consequence of the selective averaging theorem of Amrutam–Jiang.

Lemma 3.1 (Selective averaging [AJ26, Theorem 3.3 and Lemma 4.1])**.** *Let G be a torsion-free nonamenable hyperbolic group, let $t \in G$ be primitive, and let (m_j) be a strictly increasing sequence of positive integers. For every finite family $a_1, \dots, a_N \in C_r^*(G)$, every $\varepsilon > 0$, and every $J_0 \geq 1$, there is a finite convex combination*

$$\Phi = \sum_{\ell=1}^{M} \theta_\ell \alpha_{t^{m_{j_\ell}}}, \qquad j_\ell \geq J_0, \quad \theta_\ell \geq 0, \quad \sum_{\ell=1}^{M} \theta_\ell = 1,$$

such that

$$\|\Phi(a_k) - E_{\langle t\rangle}(a_k)\| < \varepsilon \qquad (1 \leq k \leq N).$$

The same assertion holds with t replaced by t^{-1}.

The cited theorem gives equal-weight averages. The convex formulation above suffices here. Appendix A recalls its proof.

Both Φ and $E_{\langle t\rangle}$ are unital completely positive: the first is a convex combination of $*$-automorphisms, and the second is a conditional expectation. Hence, for every unital C^*-algebra D, their minimal tensor lifts $\Phi \otimes \mathrm{id}_D$ and $E_{\langle t\rangle} \otimes \mathrm{id}_D$ are unital completely positive and contractive. If $a \in C_r^*(G) \otimes_{\min} D$ and $p = \sum_{k=1}^{N} a_k \otimes d_k$, then

$$\begin{aligned} &\|(\Phi \otimes \mathrm{id}_D)(a) - (E_{\langle t\rangle} \otimes \mathrm{id}_D)(a)\| \\ &\qquad \leq 2\|a - p\| + \sum_{k=1}^{N} \|\Phi(a_k) - E_{\langle t\rangle}(a_k)\|\, \|d_k\|. \end{aligned} \tag{3.1}$$

Thus one simultaneous estimate on finitely many factor elements gives the required norm approximation for each fixed tensor-product element.

Proposition 3.2. *Let $\Gamma = \Gamma_1 \times \cdots \times \Gamma_n$ be as in Theorem A. If $B \in \mathrm{Sub}(C_r^*(\Gamma))$ is boomerang, then*

$$E_{\langle t\rangle}^{(i)}(B) \subseteq B$$

for every i and every primitive $t \in \Gamma_i$.

Proof. Fix $a \in B$ and embed t in coordinate i, writing $\widetilde{t}$ for the resulting element of Γ. Choose a return sequence (m_j) for $\widetilde{t}$ at B. Equation (2.2) gives

$$d(\alpha_{\widetilde{t}^{-m_j}}(a), B) \longrightarrow 0.$$

Fix $\varepsilon > 0$ and choose J_0 so that these distances are below ε for $j \geq J_0$.

Identify A with $C_r^*(\Gamma_i) \otimes_{\min} D$, where D is the tensor product of the other factors. Choose $p = \sum_{k=1}^{N} a_k \otimes d_k$ with $\|a - p\| < \varepsilon$, and put $M_0 = \sum_k \|d_k\|$. Apply Lemma 3.1 to t^{-1} and the tail $j \geq J_0$, with error $\varepsilon/(1 + M_0)$ on each a_k. The resulting map $\Psi = \Phi \otimes \mathrm{id}_D$ is a convex combination of the conjugations $\alpha_{t^{-m_j}}$ from that tail. By (3.1),

$$\|\Psi(a) - E^{(i)}_{\langle t \rangle}(a)\| < 3\varepsilon.$$

Every summand of $\Psi(a)$ is within ε of B. Since B is convex, $d(\Psi(a), B) \leq \varepsilon$. It follows that $d(E^{(i)}_{\langle t \rangle}(a), B) < 4\varepsilon$. Letting $\varepsilon \to 0$ and using closedness of B proves the assertion. □

Corollary 3.3. *Under the hypotheses of Proposition 3.2, $\mathbb{E}_{\tau_0^i}(B) \subseteq B$ for every i. Moreover,*

$$E_{C_\Gamma(w)}(B) \subseteq B \qquad (w \in \Gamma).$$

If $J \subseteq \{1, \ldots, n\}$ and $t_i \in \Gamma_i$ is primitive for $i \in J$, then the expectation onto

$$\Lambda = \prod_{i \in J} \langle t_i \rangle \times \prod_{i \notin J} \{e\}$$

also preserves B.

Proof. Choose a primitive $u \in \Gamma_i$ and a free partner h. Then $v = huh^{-1}$ is primitive and $\langle u \rangle \cap \langle v \rangle = \{e\}$. Equation (2.1) gives

$$\mathbb{E}_{\tau_0^i} = E^{(i)}_{\langle u \rangle} E^{(i)}_{\langle v \rangle}. \tag{3.2}$$

Proposition 3.2 proves the first assertion.

For $w = (w_1, \ldots, w_n)$,

$$C_\Gamma(w) = \prod_{i=1}^{n} C_{\Gamma_i}(w_i).$$

If $w_i \neq e$, the corresponding centralizer is generated by a primitive element; if $w_i = e$, it is the whole factor. Thus $E_{C_\Gamma(w)}$ is a composition of coordinate cyclic expectations and identity maps. Finally,

$$E_\Lambda = \left(\prod_{i \in J} E^{(i)}_{\langle t_i \rangle} \right) \left(\prod_{i \notin J} \mathbb{E}_{\tau_0^i} \right). \tag{3.3}$$

All these maps commute by (2.1). □

4. Mixed Fourier coefficients

The coefficient argument begins with the reduced-word calculation in [AJ26, Lemma 2.3] and [FOT26, Lemma 6.6].

Lemma 4.1. *Let t, q freely generate a free subgroup of a group G. For $r, k, l \in \mathbb{Z} \setminus \{0\}$, put*

$$w = t^r q t^r q^{-1}, \qquad z = t^k q t^l q^{-1}.$$

Then $z \in C_G(w)$ if and only if $k = l = r$.

Proof. The products wz and zw are reduced alternating words in $\langle t \rangle * \langle q \rangle$. Equality of their first t-syllables gives $k = r$, and equality of their next t-syllables gives $l = r$. The converse is immediate. Both products lie in the embedded free subgroup, so this comparison also tests commutation in G. □

Fix a nonempty $J \subseteq \{1, \dots, n\}$, primitive $t_i \in \Gamma_i$, and nonzero integers r_i for $i \in J$. Set

$$\Lambda = \prod_{i \in J} \langle t_i \rangle \times \prod_{i \notin J} \{e\}.$$

For $\boldsymbol{k} = (k_i)_{i \in J} \in \mathbb{Z}^J$, write $t^{\boldsymbol{k}}$ for the element with coordinates $t_i^{k_i}$ on J and identity elsewhere. Put $g = t^{\boldsymbol{r}}$. Choose $q_i \in \Gamma_i$ free from t_i, and let q have coordinates q_i on J and identity elsewhere.

Lemma 4.2. *Let $x \in C_r^*(\Lambda)$ and $c = \widehat{x}(g)$, and put $w = gqgq^{-1}$. If $\mathbb{E}_{\tau_0^i}(x) = 0$ for every $i \in J$, then*

$$E_{C_\Gamma(w)}(x \lambda_q x \lambda_q^*) = c^2 \lambda_w. \tag{4.1}$$

For every $x \in C_r^*(\Lambda)$, without the preceding vanishing assumption,

$$E_\Lambda(\lambda_w \lambda_q x^* \lambda_q^*) = \overline{c}\, \lambda_g. \tag{4.2}$$

Proof. First let

$$p = \sum_{\boldsymbol{k} \in (\mathbb{Z} \setminus \{0\})^J} p_{\boldsymbol{k}} \lambda_{t^{\boldsymbol{k}}}$$

have finite support. In $p \lambda_q p \lambda_q^*$, the ordered pair $(\boldsymbol{k}, \boldsymbol{l})$ contributes $p_{\boldsymbol{k}} p_{\boldsymbol{l}}$ at the tuple with coordinates $t_i^{k_i} q_i t_i^{l_i} q_i^{-1}$ on J. Such a tuple belongs to $C_\Gamma(w)$ exactly when each coordinate commutes with $t_i^{r_i} q_i t_i^{r_i} q_i^{-1}$. Lemma 4.1 forces $\boldsymbol{k} = \boldsymbol{l} = \boldsymbol{r}$. Consequently

$$E_{C_\Gamma(w)}(p \lambda_q p \lambda_q^*) = p_{\boldsymbol{r}}^2 \lambda_w.$$

To pass from polynomials to x, define

$$Q_J = \prod_{i \in J} (\mathrm{id} - \mathbb{E}_{\tau_0^i}). \tag{4.3}$$

The coordinate trace maps commute. The hypotheses give $Q_J x = x$, and $\|Q_J\| \leq 2^{|J|}$. Take $p_m \in \mathbb{C}[\Lambda]$ converging to x in norm and set $z_m = Q_J p_m$. Then $z_m \to x$ in norm, and each z_m has only nonzero exponents in every coordinate of J. The estimate

$$\|z_m \lambda_q z_m \lambda_q^* - x \lambda_q x \lambda_q^*\| \leq (\|z_m\| + \|x\|) \|z_m - x\|$$

and continuity of Fourier coefficients prove (4.1). This passage uses polynomial density, without requiring convergence of Fourier partial sums.

For the second identity, freeness gives $\Lambda \cap q \Lambda q^{-1} = \{e\}$. Checking group unitaries and using norm density yields

$$E_\Lambda(\lambda_q y \lambda_q^*) = \tau_0(y) 1 \qquad (y \in C_r^*(\Lambda)).$$

Since $\lambda_w = \lambda_g \lambda_q \lambda_g \lambda_q^*$, bimodularity of E_Λ gives

$$\begin{aligned} E_\Lambda(\lambda_w \lambda_q x^* \lambda_q^*) &= \lambda_g E_\Lambda(\lambda_q (\lambda_g x^*) \lambda_q^*) \\ &= \tau_0(\lambda_g x^*) \lambda_g = \overline{c}\, \lambda_g. \end{aligned}$$

Only membership of the bimodule factor λ_g in $C_r^*(\Lambda)$ is used here; its membership in B is the conclusion still to be proved. □

Remark 4.3. The coordinate vanishing assumptions in (4.1) remove the case $k_i = l_i = 0$, which otherwise contributes an identity in coordinate i. The coefficients of p need not factor across coordinates. The surviving coefficient is the square of the single coefficient $p_{\boldsymbol{r}}$.

5. Proof of the main statements

Proof of Theorem A. Put $H_B = \{g \in \Gamma : \lambda_g \in B\}$. This is a subgroup and $C_r^*(H_B) \subseteq B$. We prove that

$$a \in B, \quad \widehat{a}(g) \neq 0 \quad \Longrightarrow \quad \lambda_g \in B. \tag{5.1}$$

The case $g = e$ follows from unitality. Otherwise let $J = \{i : g_i \neq e\}$. For each $i \in J$, write $g_i = t_i^{r_i}$ with t_i primitive and $r_i \neq 0$. Form Λ and Q_J as in Section 4, and set

$$x = Q_J E_\Lambda(a). \tag{5.2}$$

Corollary 3.3 implies $x \in B \cap C_r^*(\Lambda)$. Moreover $\mathbb{E}_{\tau_0^i}(x) = 0$ for every $i \in J$, and

$$c := \widehat{x}(g) = \widehat{a}(g) \neq 0.$$

Indeed, E_Λ retains the coefficient at g, while $\mathbb{E}_{\tau_0^i}$ annihilates that coefficient for each $i \in J$.

Choose h_i free from t_i for every $i \in J$, and let $h \in \Gamma$ have these coordinates and identity elsewhere. Choose a single return sequence (m_j) for h at B. Put

$$q_j = h^{-m_j}, \qquad w_j = g q_j g q_j^{-1}, \qquad \delta_j = d(\lambda_{q_j} x \lambda_{q_j}^*, B).$$

Equation (2.2) gives $\delta_j \to 0$. For every j and $i \in J$, the pair $t_i, h_i^{-m_j}$ is free: substitution into a reduced word in these two elements leaves a nontrivial reduced word in t_i, h_i. Thus Lemma 4.2 applies with $q = q_j$.

Let $K = \|x\|$. Since $x \in B$,

$$d(x\lambda_{q_j} x \lambda_{q_j}^*, B) \leq K\delta_j.$$

The expectation $E_{C_\Gamma(w_j)}$ is contractive and preserves B by Corollary 3.3. Equation (4.1) therefore gives

$$\varepsilon_j := d(\lambda_{w_j}, B) \leq \frac{K}{|c|^2}\delta_j \longrightarrow 0. \tag{5.3}$$

Adjoints preserve distance to B. Using (2.4), (2.3), and (4.2), we obtain

$$\begin{aligned} |c|\, d(\lambda_g, B) &= d(E_\Lambda(\lambda_{w_j}\lambda_{q_j} x^* \lambda_{q_j}^*), B) \\ &\leq \delta_j + K\varepsilon_j + \varepsilon_j\delta_j \longrightarrow 0. \end{aligned}$$

This proves (5.1). The same return sequence controls all active coordinates throughout.

For $a \in B$, (5.1) shows that its Fourier coefficients vanish outside H_B. Hence $a - E_{H_B}(a)$ has all Fourier coefficients zero. Such an element vanishes on δ_e and, by commutation with the right regular representation, on every δ_g. It is therefore zero. Thus $a = E_{H_B}(a) \in C_r^*(H_B)$, proving the opposite inclusion. □

Proof of Corollary B. Let ν be an invariant random subalgebra of $C_r^*(\Gamma)$. By [FOT26, Proposition 6.3], almost every B is boomerang. This is Poincaré recurrence applied to a countable base of $\mathrm{Sub}(C_r^*(\Gamma))$, followed by a countable intersection over $s \in \Gamma$. Theorem A gives $\nu(I_\Gamma) = 1$, where

$$I_\Gamma = \iota_r(\mathrm{Sub}(\Gamma)).$$

For completeness, the subgroup recovery map is Borel, since

$$\{B : g \in H_B\} = \{B : d(\lambda_g, B) = 0\}$$

is Wijsman closed for each g. It is equivariant because $H_{\lambda_s B \lambda_s^*} = s H_B s^{-1}$. The map ι_r is a Borel isomorphism onto I_Γ [FOT26, Theorem 2.13]. Also $H_{C_r^*(H)} = H$: if $\lambda_g \in C_r^*(H)$, applying E_H shows that $g \in H$.

Set $\eta = (B \mapsto H_B)_*\nu$. Equivariance makes η an IRS, and Theorem A yields $(\iota_r)_*\eta = \nu$. Applying the recovery map to any other representing measure proves uniqueness. Both pushforward maps are affine. Their equivariant Borel inverse correspondence identifies invariant measurable sets, and hence preserves and reflects ergodicity. □

6. Restricted direct sums and scope

Corollary 6.1. *Let $\Gamma = \bigoplus_{i\in\mathbb{N}} \Gamma_i$, where each Γ_i is a torsion-free nonamenable hyperbolic group. Every boomerang unital subalgebra of $C_r^*(\Gamma)$ is a subgroup algebra. Every invariant random subalgebra is geometric, with the unique IRS given by the map $B \mapsto H_B$.*

Proof. For each i,

$$C_r^*(\Gamma) = C_r^*(\Gamma_i) \otimes_{\min} C_r^*\left(\bigoplus_{k\neq i} \Gamma_k\right).$$

The proof of Proposition 3.2 applies to this decomposition. Equation (3.2) then proves preservation of all coordinate trace expectations.

Fix a finite set $J \subseteq \mathbb{N}$ and put $\Gamma_J = \prod_{i\in J} \Gamma_i$. Choose finite sets F_m increasing to $\mathbb{N} \setminus J$. The maps $\prod_{i\in F_m} \mathbb{E}_{\tau_0^i}$ converge in norm on each element to E_{Γ_J}. They eventually agree with E_{Γ_J} on each group polynomial, since every element of Γ has finite coordinate support. Contractivity gives convergence on all of $C_r^*(\Gamma)$. Closedness of B implies $E_{\Gamma_J}(B) \subseteq B$.

For a selected nonidentity g, its set J of nontrivial coordinates is finite. Apply E_{Γ_J}, the cyclic expectations on J, and Q_J to construct (5.2). The coefficient calculation and the return sequence use only these coordinates. Each expectation $E_{C_\Gamma(w_j)}$ is a composition of the finitely many cyclic expectations on J, with the identity on the remaining coordinates. The proof of (5.1) is therefore unchanged. Subgroup recovery and the proof for invariant measures apply because Γ is countable. □

The factor properties used in the argument can be stated independently of hyperbolicity:

(i) every nonidentity element has an infinite cyclic centralizer with a primitive generator;
(ii) every primitive element admits a free partner;
(iii) simultaneous selective averaging onto the primitive cyclic subgroup is available along every subsequence of powers, in the form of Lemma 3.1.

The same proof applies to finite products and countable restricted direct sums of nontrivial countable groups with these properties. The tensor estimate (3.1) requires no nuclearity or approximation property.

These hypotheses have not been established here for general torsion-free acylindrically hyperbolic groups. In particular, cyclic centralizers are not available for all their nonidentity elements. No assertion is made for unrestricted infinite direct products.

The classification reduces the study of invariant random subalgebras for these groups to the study of their IRSs. It does not enumerate those IRSs, nor does it resolve the general questions relating C^*-ISR, von Neumann ISR, and C^*-simplicity in [AJ26, Questions 6.11–6.12].

Appendix A. Selective averaging along subsequences

We record the convex averaging argument used in Lemma 3.1. It is adapted from the boundary-state argument of [AJ26, Lemma 3.1 and Proposition 3.2].

Lemma A.1. *Let G be a countable group acting continuously on a compact metrizable space X. Suppose that $t \in G$ has north–south dynamics in both time directions, with fixed points t^+, t^-, and*

$$s\{t^+, t^-\} \cap \{t^+, t^-\} = \varnothing \qquad (s \notin \langle t \rangle).$$

Then the conclusion of Lemma 3.1 holds for t.

Proof. Put $F = \{t^+, t^-\}$. Extend a state φ on $C_r^*(G)$ to a state ω on $C(X) \rtimes_r G$, using the canonical unital inclusion. If $\rho = \omega|_{C(X)}$, then

$$(\omega \circ \alpha_{t^{-n}})(f) = \int_X f(t^n x)\, d\rho(x) \qquad (f \in C(X)).$$

North–south dynamics and dominated convergence show that these restrictions converge to a probability measure supported on F.

Let ω_∞ be a weak-* cluster point of the states $\omega \circ \alpha_{t^{-n}}$. For $s \notin \langle t \rangle$, choose $f \in C(X)$ with $0 \le f \le 1$, $f|_F = 1$, and $f|_{sF} = 0$. Cauchy–Schwarz gives

$$|\omega_\infty(f\lambda_s)|^2 \le \omega_\infty(f)\omega_\infty(\lambda_s^* f \lambda_s) = 0,$$
$$|\omega_\infty((1-f)\lambda_s)|^2 \le \omega_\infty(1-f)\omega_\infty(\lambda_s^*(1-f)\lambda_s) = 0.$$

Thus $\omega_\infty(\lambda_s) = 0$. Conjugation by powers of t fixes $C_r^*(\langle t \rangle)$ pointwise. Consequently, the restriction of every such cluster state to $C_r^*(G)$ is $\varphi \circ E_{\langle t \rangle}$: the functionals agree on group unitaries, and hence on all of $C_r^*(G)$ by norm density. Weak-* compactness therefore gives

$$(\omega \circ \alpha_{t^{-n}})|_{C_r^*(G)} = \varphi \circ \alpha_{t^{-n}} \xrightarrow{w^*} \varphi \circ E_{\langle t \rangle}.$$

This argument establishes convergence of the restricted states; convergence of the full states on the crossed product is not needed.

Every bounded linear functional is a linear combination of states. Therefore $\alpha_{t^{-n}}(a)$ converges to $E_{\langle t \rangle}(a)$ in the Banach weak topology for every $a \in C_r^*(G)$. For a finite family, the corresponding tuple converges weakly in a finite direct sum of copies of $C_r^*(G)$. By Hahn–Banach separation, its limit belongs to the norm-closed convex hull of every subsequence tail. This gives one common convex combination for the family. Interchanging t and t^{-1} proves the assertion in both directions. □

For primitive elements of torsion-free nonamenable hyperbolic groups, the boundary separation hypothesis is [AJ26, Lemma 4.1]. Applying the lemma to the Gromov boundary therefore gives the form of selective averaging used in the proof.

Preparation

Internal calculations and finite reduced-word checks were used during preparation. The bibliography identifies the versions of the cited works used in the argument.

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